Government Healthcare Spending and Life Expectancy

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1 Introduction

Does allocating a larger share of a government's fiscal budget to healthcare increase population longevity? In global public policy discussions, an intuitive positive association is often presumed: nations with higher public health spending shares enjoy longer life expectancies. However, from an econometric perspective, observing that high-spending nations live longer does not establish that public budgetary reallocations causally generate those longevity gains. Cross-national comparisons are profoundly vulnerable to omitted-variable bias, reverse causality, and unobserved institutional heterogeneity.

This study develops an incremental empirical investigation across 181 nations from 2000 to 2015, moving step-by-step from raw cross-sectional correlations to panel identification with Two-Way Fixed Effects, nonlinear threshold tests, dynamic lag horizons, and health shock diagnostics.

2 Data Pre-Processing

2.1 Documenting the Primary Explanatory Variable

The core explanatory variable, **Total expenditure**, is defined strictly following the official World Bank guidelines:

$$\text{Total expenditure} \equiv \text{Domestic general government expenditure on health as a percentage of general government expenditure} \tag{1}$$

It is essential to emphasize what this variable measures: it captures fiscal prioritization; the proportion of the public budget envelope allocated to healthcare. It does *not* measure real per-capita monetary spending. In a low-income nation, a 15% budget share may represent less than $20 per person annually, whereas an 8% share in an advanced economy represents thousands of dollars per citizen.

2.2 The Data Harmonization Pipeline

The original compilation suffered from crowdsourced discrepancies, unreconciled country names, and inaccurate developmental classifications. To establish an authoritative panel, primary sources

were integrated:

1. Official WHO life expectancy records.
2. World Bank constant USD GDP per capita series.
3. World Bank health expenditure shares.

Country identifiers were harmonized across 28 nation pairs, non-positive GDP records removed, and developmental classifications corrected (specifically France and Canada, mislabeled as 'Developing' in the raw file, reclassified as 'Developed'). The resulting clean panel contains 2,702 balanced country-year records across 181 countries spanning 2000-2015.

3 Motivating Evidence & Correlation Structure

Bivariate Pearson correlations highlight the underlying web of macroeconomic co-movements. While life expectancy exhibits a moderate positive association with health expenditure share ($r = +0.497$), it is far more powerfully correlated with national income ($r = +0.797$ with ln(GDPpc)) and education ($r = +0.770$ with Schooling). Crucially, the health expenditure share is itself substantially correlated with ln(GDPpc) ($r = +0.475$) and Schooling ($r = +0.503$). Wealthier, more educated societies consistently dedicate larger fractions of their public budgets to healthcare. Consequently, unadjusted regressions between health spending and longevity are inevitably contaminated by the broader benefits of general national modernization.

4 Simpson's Paradox & The Confounding Role of National Wealth

To provide a vivid, intuitive demonstration of how omitted national wealth can distort empirical findings, we examined national alcohol consumption (pure litres per capita per year). In naive cross-country pooled data, drinking alcohol exhibits a strong, statistically significant positive association with longevity ($\beta = +0.8651$, $p < 0.0001$).

This finding is a textbook illustration of Simpson's Paradox: high-income nations drink nearly three times as much alcohol as developing nations, but their superior living standards, nutrition, and advanced medical infrastructure afford them vastly greater longevity. Once developing countries were stratified into homogeneous GDP per capita terciles, the relationship inverted to strictly negative in every single tier: Poorest ($\beta = -0.5241$, $p < 0.0001$), Middle ($\beta = -0.2618$, $p = 0.005$), Richest developing ($\beta = -0.2277$, $p < 0.0001$), and Two-Way Fixed Effects Panel Model ($\beta_{\text{TWFE}} = -0.1458$, $p = 0.0040$).

This establishes the central econometric warning of our investigation: national income is a master confounder capable of masking deleterious health factors as positive.

5 Progressive Econometric Modeling

The analysis starts with contemporaneous pooled OLS using the same 2,702 country-year observations. The preferred baseline excludes Adult Mortality and controls for log GDP per

capita, Schooling, and development status:

$$LE_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 \ln(\text{GDPpc}_{it}) + \beta_3 \text{Schooling}_{it} + \beta_4 \text{Developed}_i + u_{it} \tag{2}$$

Table 1: Baseline OLS Regression Comparisons

Baseline specification	**Coeff.**	**OLS SE**	**p-value**	**95% CI**	R^2
Without Adult Mortality	0.1927	0.0243	< 0.001	[0.1450, 0.2404]	0.6950
With Adult Mortality	0.1484	0.0203	< 0.001	[0.1086, 0.1882]	0.7889

6 Dynamic Lags & The Rejection of Diminishing Returns

To test diminishing returns, we estimated a piecewise linear model fixing a threshold at sample median spending (8.76%). The estimated slope change is small and negative (≈ -0.075), but far from statistically significant ($p = 0.76$). Consequently, we find no empirical evidence of diminishing marginal returns.

Table 2: Dynamic Lag Horizon Estimates

Lag Horizon	**Slope Below Knot**	**Slope Change**	**p-value (Cluster SE)**	**Net Slope Above**
Contemporaneous ($L = 0$)	+0.2456	−0.0767	0.7456	+0.1689 ($p = 0.074$)
1-Year Lag ($L = 1$)	+0.2321	−0.0749	0.7562	+0.1572 ($p = 0.115$)
3-Year Lag ($L = 3$)	+0.1860	−0.0386	0.8783	+0.1474 ($p = 0.174$)

7 Reverse Causality and Panel Identification

A primary theoretical threat to causal identification is reverse causality. Our initial proposal raised the concern that a severe health crisis could reduce life expectancy while simultaneously prompting governments to increase health expenditure in response. Such a mechanism could conceal a true positive effect by introducing a downward bias in the regression coefficient.

To investigate this possibility, we defined a health crisis shock as an annual decline in national life expectancy of at least one year ($\Delta\text{LifeExp} \leq -1.0$). Across the entire 2000–2015 panel, we identified 26 such acute crisis episodes, including Haiti in 2010 following the catastrophic earthquake and Syria in 2012 during the outbreak of civil war. In the shock year, national life expectancy declines by an average of approximately 2.5 years. However, an examination of the fiscal response reveals no clear or systematic reaction in government health expenditure. The estimated annual change in spending following a shock is an increase of approximately 0.2 percentage points, which is statistically insignificant ($\beta = +0.2008$, $p = 0.465$). The corresponding

estimate when using lagged life expectancy declines is also statistically insignificant ($\beta = -0.0225$, $p = 0.724$).

This motivated our final diagnostic test: whether controlling for health shocks or excluding crisis episodes would reveal a positive spending effect. The empirical contrast is striking. Estimations without fixed effects, based on pooled cross-sectional models, show a positive association of approximately 0.2 years of life expectancy per percentage point of health expenditure ($\beta \approx +0.180$ to $+0.249$, $p < 0.05$). Neither controlling for lagged shocks, controlling for HIV prevalence, nor excluding shock-affected observations substantially changes this magnitude.

However, once country and year fixed effects are introduced, thereby comparing each country with itself over time, the estimated effect falls to virtually zero across all specifications. The coefficients are $\beta = -0.0085$ in the baseline two-way fixed effects (TWFE) model, $\beta = -0.0059$ with shock and HIV controls, $\beta = -0.0068$ after excluding shock-affected observations, and $\beta = -0.0007$ after excluding countries that experienced shocks entirely. These findings indicate that the positive association observed in pooled regressions primarily reflects broad cross-sectional differences in development between countries, rather than changes in health expenditure within individual countries over time.

In conclusion, our empirical tests find no evidence of diminishing returns, no detectable delayed multi-year effect, and no statistically significant evidence of reverse causality. Importantly, these findings do not imply that public health expenditure is ineffective. Our primary measure captures the share of the government budget allocated to health rather than actual health expenditure per capita. Furthermore, our identification strategy relies on a finite sample containing only 26 acute health shock episodes.

8 Theoretical Context: Technology Spillovers

A comprehensive theoretical discussion must address cross-country medical technology spillovers (such as vaccines, antivirals, and modern pharmaceuticals developed in advanced economies benefiting poorer nations). Advanced nations finance the massive fixed capital and scientific R&D costs required to invent medical technologies. Once discovered, these innovations diffuse globally at very low marginal cost, often distributed through international consortia (such as the WHO, Gavi, and the Global Fund).

As a result, developing nations achieve dramatic improvements in population longevity (e.g., eradicating polio, controlling measles, deploying antiretrovirals) without needing to expand their domestic health budget shares. This mechanism explains why cross-group slope comparisons are difficult to interpret cleanly: the observed return to domestic healthcare spending in developing nations is confounded and compressed by external global technology spillovers.

Some possible explanations are:

- Developed countries already have strong health infrastructure and are close to the medical frontier. Extra spending therefore tends to produce small and slow gains in life expectancy. Furthermore, our dataset covers only a 15-year period, which is relatively short for capturing the long-term effects of healthcare spending on life expectancy and may therefore limit our ability to identify its full impact.

- Developing countries have much more room to improve, and progress can be fast. But they often lack the funds to invest. In many cases, wealthier countries and international donors provide part of the money, so life expectancy can improve even without a large increase in the government's own health budget. Our model does not capture this: it only measures domestic government spending as a share of the budget, not foreign aid.

9 Conclusion

9.1 The Empirical Reality: A Non-Causal Link

The naive cross-sectional association between government health expenditure share and life expectancy (+0.84 years per percentage point) is non-causal. Once national wealth (log GDP) and education (`Schooling`) are controlled for, this positive effect completely collapses. When evaluating within-country changes over time via **Two-Way Fixed Effects (TWFE)**, the estimated impact falls to precisely zero ($\beta = -0.032$, $p = 0.224$). Our diagnostic tests confirm:

- **Diminishing returns**: A piecewise threshold at median spending (8.76%) shows no significant flattening ($p = 0.76$).
- **Delayed effect**: Dynamic lags (1-year and 3-year horizons) effects could not be observed.
- **Reverse causality**: Acute health crises (ΔLifeExp ≤ -1.0 year across 26 historical shock episodes) do not trigger compensatory spending spikes.

9.2 Theoretical Mechanisms & Technology Spillovers

Two structural mechanisms explain why within-country budgetary reallocations fail to move aggregate longevity:

1. **Budget Share vs. Real Purchasing Power**: `Total expenditure` measures the political priority of health (% of the public budget), not real dollars per citizen. A 15% allocation in a poor country yields under $20 per person, whereas an 8% share in a rich nation provides thousands. Real resources are dictated by GDP, which absorbs the cross-sectional explanatory power.
2. **Cross-Country Medical Spillovers**: Advanced nations finance the fixed R&D costs of medical breakthroughs (vaccines, antivirals, antibiotics). These technologies diffuse globally at near-zero marginal cost through agencies like the WHO, Gavi, and the Global Fund. Consequently, developing nations achieve massive longevity gains without needing to increase their domestic health budget shares, obscuring cross-group slope comparisons.